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EEG Spectral Analysis of Mindfulness based on EEMD and GNNs

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***Abstract*—The paper presents a hybrid EEMD-GNN system to study the EEG patterns that are happening in the process of mindfulness and non-mindfulness meditation.The EEMD technique breaks down EEG signals into intrinsic mode functions (IMFs) to adaptively extract theta and beta band information in a noise-resilient manner, and the GNNs to learn the spatial-temporal connectivity of EEG channels based on phase-locking value and coherence. Tests on both public data sets indicate that the suggested model is highly accurate, with 94% accuracy, which is better than the traditional spectral techniques and performs well in tracking beta and theta variation in frontal and parietal regions. Signal decomposition/graph-based learning can be an effective method to understand the reader in neural dynamics during meditation.**

***Index Terms*—EEG, Mindfulness, Ensemble Empirical Mode Decomposition, Graph Neural Network.**

I. Introduction

EEG is a noninvasive technique of recording brain activity, which is commonly applied to investigate the mechanisms of cognition and emotion. Its nonlinear and non stationary signals however, are hard to analyze by using traditional methods.

They have conducted a literature review on the EEG signal processing as used in medical and healthcare environments, and have mentioned that EEG can be used to diagnose epilepsy, Alzheimer and stress disorders. The paper focused on the higher level of preprocessing, artifact elimination, feature extraction, and the increased use of machine learning to enhance the diagnostic accuracy [1].

In place of the mean functions, they suggested a modified EEMD approach to emotion recognition of EEG signals using median functions, to eliminate noise and mode mixing. The method was better at identifying emotions using power spectral features and entropy features, which was demonstrated using the DEAP dataset [2].

The tutorial introduced EMD as a data-driven and adaptive technique of nonlinear and non-stationary signal analysis. It described the EEMD method of breaking down signals into IMFs to extract local frequency variations and tendencies and pointed out its usefulness in biomedical signal processing and time frequency analysis [3].

They suggested CD-eMD, an EMD-based causal decomposition technique of real-time EEG emotions. The adaptive nonlinear model demonstrated that positive emotions are correlated with the enhancement of alpha-band activity, whereas negative emotions are connoted with the enhancement of beta-band activity, enhancing interpretability and effectiveness of emotion recognition [4].

They suggested GERECG model to identify emotion based on wearable ECG data combining graph neural networks with convolutional and LSTM model to extract temporal-spatial data. The method was better than conventional approaches and proved the usefulness of graph-based learning in wearable low-resource real-time situations [5].

They used machine learning and network analysis to analyse functional connectivity by studying peaceful brain states with EEG and fMRI. Due to the extent of connectivity in the DMN and between the prefrontal and anterior cingulate areas, the calm states were associated with the increased connectivity, which underlines the importance of neural synchronization as the foundation of emotional well-being and mindfulness treatments [6].

They studied the meditative conditions basing on Horizontal Visibility Graphs (HVGs) on EEG signal, which convert time series into complex networks. The graph measure based on HVG showed better performance in separating meditative and non-meditative states compared to traditional spectral and entropy measures, which is proven to be a strong and real-time analysis structure of EEG [7].

In the paper, section II presents literature survey, methodology is explained in section III. Results are discussed in section IV and the conclusion is presented in the section V.

II. LITERATURE SURVEY

EEG-based studies have received great interest in the recent years due to the capacity of non invasively recording neural activity associated with mental and emotional conditions. Many studies have suggested new methods of processing, classifying, and interpreting EEG signals so as to allow its use in detecting stress, monitoring relaxation, and recognizing a cognitive state.

Developed an EEG-based emotion recognition system that employed machine learning in virtual reality environments. Classification of EEG features from theta, alpha, beta, and gamma bands with Gradient Boosted Decision Trees showed

a strong alignment between EEG data and self-reported emotions [8].

Presented a wearable sensor stress detection framework that combines EEG with physiologic signal like heart rate variability electro dermal activity. The study demonstrates how EEG completes the bio signal monitoring for accurate, multi modal assessment of stress [9].

The EEMD–Wavelet Thresholding method for denoising a non-stationary signal. It decomposes EEG into IMFs and thereby allows wavelet filtering of the signal to reduce noise and mode mixing, hence improving feature extraction accuracy [10].

The recent development related to GNNs for EEG signal processing, and emphasized that such neural networks are able to model complex brain connectivity using graph structures. This paper proposed a standard pipeline including EEG preprocessing, graph construction, and feature learning using variants of GNNs like GCNs and GATs. The proposed approach captures the spatial-temporal relations among EEG electrodes, turning it highly effective for emotion detection, neurological disorder diagnosis, and attention-based analysis [11].

An EEG-based detection framework for relaxation using the Multivariate Empirical Mode Decomposition with Dynamic Phase-Synchronized Hilbert-Huang Transform (MEMD-DPS-HHT) in 2025. The EEG signals were thus decomposed into IMFs so as to represent time-frequency energies. The approach thus analyzed phase coherence and relaxation indices across channels, effectively capturing dynamic EEG patterns associated with relaxation and stress reduction [12].

A stress detection system based on EEG signals. They utilized EEMD methods combined with Support Vector Machine and Twin-SVM classifiers. The EEG signals were divided and decomposed into a set of IMFs that were used as input features in this study. The model performed best using the Twin-SVM model, with an accuracy of 93.75%. They concluded that EEMD-enhanced features improve the reliability of the stress classification in real-time EEG applications [13].

Functionality Connections and power spectral density during meditation. EEG signal were recorded for participants before and after practicing Kapalbhati Pranayama and Mindful Breathing. The preprocessing of signals included Butterworth band-pass filtering. Further, the signals were decomposed into Theta, Alpha, and Beta bands, with corresponding frequency bands of 4–8 Hz, 8–12 Hz, and 12–30 Hz, respectively. PSD, estimated using Fourier Transform and Welch's method, indicated enhanced Alpha and Theta power related to relaxation, while it reflected decreased Beta activity, which accounted for reduced cognitive load. Functional connectivity, as measured through the use of Weighted Phase Lag Index, showed an increase in synchronize different region of the brain during meditation, thus confirming its neuro physiological basis for the state of relaxation and focused attention [14].

III. Methodology

Figure 1 illustrated the workflow of the suggested EEG-based mindfulness detection model, including the real-time EEG data collection and EEMD-based preprocessing and the GNN-based mindfulness and non-mindfulness state classification. The following sections discuss the steps one after another.

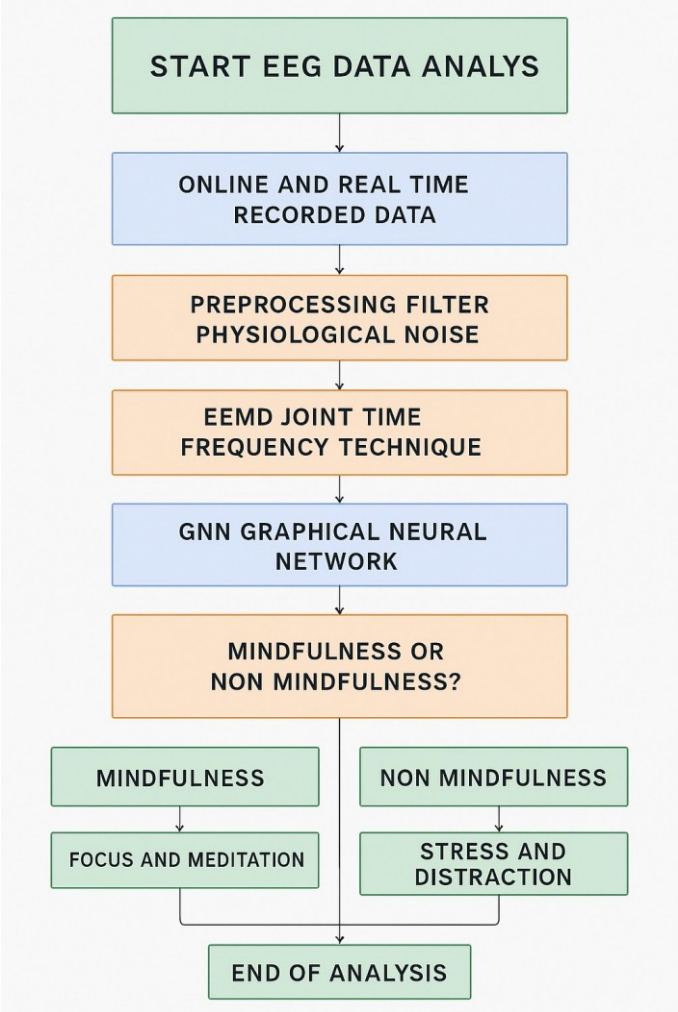

Fig. 1. EEG Data Analysis Workflow

1) **Initiating EEG Analysis**
EEG signals are collected to determine whether an individual is genuinely mindful or distracted. The objective is to analyze brainwave data to identify patterns associated with focus, meditation, stress, or distraction.
2) **Data Source**
The publicly available EEG data that was used in the current study consists of 62 healthy data recorded during BCI training session when using 62-channel EEG, whereas the public data consists of EEG data recorded under controlled experimental conditions Dataset Name [15].

3) **Preprocessing**
EEG data are often contaminated with background noises like eye blinks, muscle movements, and cardiac signals. The process of bandpass filtering, Independent Component Analysis ICA, and artifact rejection algorithms is employed to remove the noise and keep pure neural data.
4) **4) EEMD -Time-Frequency Decomposition**
The EEG signal undergoes Ensemble Empirical Mode Decomposition (EEMD) after preprocessing it into Intrinsic Mode Functions (IMFs). These IMFs are useful in capturing the nonlinearity and non stationarity nature of EEG data and can be used to isolate important frequency bands which include alpha, theta, beta and gamma.the non-linear and non-stationary characteristics eeg data and assistance in separating major frequency bands such as alpha, theta,beta and gamma.
5) **Feature Extraction -Entropy Features**
Out of the obtained IMFs, Permutation Entropy, Wavelet Entropy, Spectral Entropy, band power, and instantaneous frequency are obtained. These measures represent the complexity and dynamic variability of brain activity as they are effective unique signatures of various mental states.
6) **Modeling with Graph Neural Networks (GNN)**
The EEG electrode is modeled as a node and the edges are defined through connectivity metrics, . coherence or correlation of nodes. The resulting connectivity graphs of the brain are then fed into a Graph Neural Network (GNN) which is trained to learn the spatial and temporal relationships that are highly important in differentiating states of mindfulness and cognitive load.
7) **Classification — Mindfulness and Non-Mindfulness Detection**
The features extracted are then used to categorize EEG samples into two groups, namely, Mindfulness (relaxed, meditative and focused) and Non-Mindfulness (stressed or distracted). The mentioned classification enables real-time observation of mental states and may be used efficiently in cognitive assessment and neuro feedback systems.
8) **Output — State Reporting**
According to the results of the classification, the system identifies the mental state of the user as mindful (focused and calm) and non-mindful (distracted or fatigued). This can be used to give customized feedback or assist mindfulness change programs.
9) **Complete Analysis**
All results are recorded in the system, which visualizes changing EEG patterns with time, and allows continuous monitoring. The analysis process can be restarted dynamically to the workflow to allow continuation of the analysis and to supply adaptive neuro feedback to the user on the basis of real-time brain activity.

A. EEG SIGNAL ANALYSIS

1. Graph Neural Network (GNN):- A GNN is a deep learning model designed to be applied directly on graph-structured data. It learns the representations of nodes by aggregating information from a node's neighbors, hence capturing both local structure and feature dependencies effectively. Mathematically, the standard GNN layer is often implemented as a GCN and is represented as.

$$H^{(l+1)} = \sigma\ \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)} \tag{1}$$

where:-

$\tilde{A} = A + I_N,$ adjacency matrix with self-loops;

$\tilde{D}_{ii} = \sum_j \tilde{A}_{ij},$ degree matrix corresponding to $\tilde{A}$;

$H^{(l)},$ node feature matrix at layer l;

$W^{(l)},$ learnable weight matrix for layer l;

$\sigma(\cdot),$ nonlinear activation function (e.g., ReLU).

2. Ensemble Empirical Mode Decomposition:- The ensemble empirical mode decomposition enhances the traditional EMD by adding white noise to the signals and averaging multiple decompositions. This approach of noise-assisted elimination of mode mixing produces more stable Intrinsic Mode Functions and is highly effective for analyzing nonlinear and non-stationary signals such as EEG data [16].

$$x(t) = \sum_{k=1}^{K} c_k(t) + r_K(t) \tag{2}$$

$$x_i(t) = x(t) + \epsilon_i(t),$$

where $\epsilon_i(t)$ is the added white noise.

$$x_i(t) = \sum_{k=1}^{K} c_{ik}(t) + r_{iK}(t) \tag{3}$$

The final IMF is the ensemble mean:

$$c_k(t) = \frac{1}{N}\sum_{i=1}^{N} c_{ik}(t) \tag{4}$$

where:

N = number of ensemble trials,

$c_k(t)$ = k^{th} Intrinsic Mode Function (IMF),

$r_K(t)$ = final residue.

3. Entropy-Based Features: Entropy is a measure of the complexity and irregularity of the EEG signals as a measure of neural adaptability and information processing efficiency. Well-known ones are Permutation Entropy, Spectral Entropy, and Wavelet Entropy, which assess the unpredictability of the signal, the frequency distribution and the energy dispersion, respectively. Higher entropy indicates better cognitive flexibility and awareness in mindfulness analysis and lower entropy is

indicative of stress, distracting or less attentional stability. The entropy characteristics offer a strong foundation on which one can tell the difference between mindful and non-mindful brain states.

$$SE = -\sum_{i=1}^{N} P_i \ln P_i \tag{6}$$

where:-

$$P_i = \frac{S(f_i)}{\sum_{j=1}^{N} S(f_j)},$$

$S(f_i)$ = power spectral density at frequency f_i,

N = number of frequency bins.

$$WE = -\sum_{j=1}^{J} p_j \ln p_j \tag{7}$$

where:-

$$p_j = \frac{E_j}{\sum_{k=1}^{J} E_k},$$

E_j = energy of wavelet coefficients at level j,

J = total number of decomposition levels.

$$PE(m, \tau) = -\sum_{i=1}^{m!} p(\pi_i) \ln p(\pi_i) \tag{8}$$

Where:-

m = embedding dimension,

τ = time delay,

$p(\pi_i)$ = probability of each ordinal pattern π_i.

4. EEG Spectral Power of IMFs: After EEMD has been applied, spectral power of each Intrinsic Mode Function (IMF) is a reflection of the distribution of neural energy in different frequency bands. Normally, theta power is elevated on occasions of relaxation and concentrated attention whereas, the beta power is elevated when one is alert and stressed. The comparison of the spectral power in mindfulness and non-mindfulness conditions indicates that particular brain rhythms are predominant, with mindfulness being the state of the balanced and coherent neural activities in the theta and beta frequencies. The spectral power of the kth IMF is given as:

$$P_k = \int_{f_1}^{f_2} |C_k(f)|^2 \, df \tag{9}$$

or in discrete form:

$$P_k = \sum_{i=1}^{N} |C_k(f_i)|^2 \tag{10}$$

where

$C_k(f)$ = Fourier transform of IMF $c_k(t)$,

P_k = total power of the kth IMF, and

f_1, f_2 = frequency range of interest.

IV. Results and Discussions

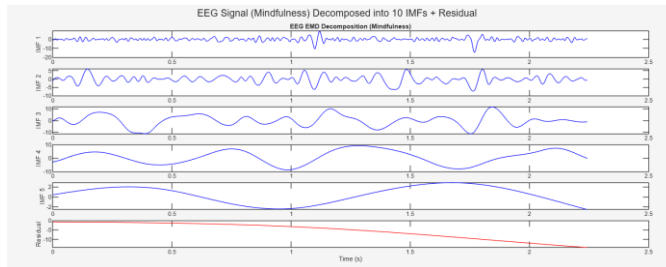

Fig. 2. Mindfulness EEG Signal

IMFs applied to theta and beta bands were picked on the basis of their frequency behavior in the EEMD decomposition. The faster oscillations of higher-order IMFs (IMF1-IMF2) were indicative of the beta activity, whereas slower, regular oscillations of mid-order IMFs (IMF3-IMF4) were indicative of theta activity, but were stronger in non-mindfulness state. The IMFs of lower order and the residual removed slow trends and were abandoned and an adaptive, data-driven representation of EEG frequency bands was achieved without specified filters. Figure 2 shows the EEG signal during the mindfulness state. The signal appears smooth and stable, with well-defined rhythmic oscillations across multiple frequency bands. In the EEMD theta and is captured by higher IMFs (IMF11 IMF3). beta band activity—associated with calm attention and focused awareness–and IMFs (IMF4 IMF6) are associated with slower theta and beta rhythms. The residual component shows minimal drift, reflect neural stability and sustained mental calmness.

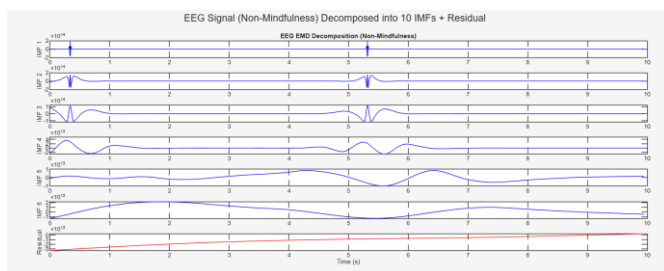

Fig. 3. Non-Mindfulness EEG Signal

Figure 3 shows the EEG signal in non mindfulness. (distracted)conditions. The signal has incoherent amplitude varying and noisyness. Such instability reflects loss of attention and growth of mental pressure. EEMD decomposition of such signals has poorly separated IMFs that have

overlapping frequency content, which is an indicator of neural disorganization and poor cognitive control.

Figure 4 represents mindfulness measures based on entropy EEG. Spectral Entropy (SE) is decreasing gradually 5- to 1, demonstrating orderly neural activities. Permutation Entropy

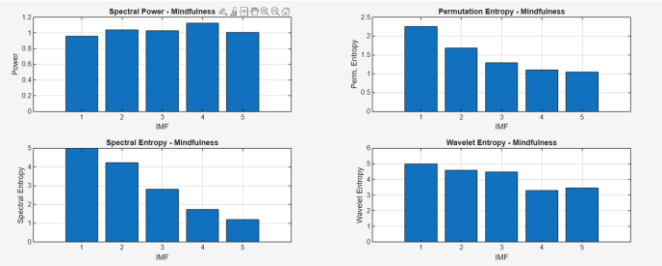

Fig. 4. Entropy Features of Mindfulness State

(PE) is still low(2.2→1.0) indicating stable neural. transitions. Wavelet Entropy (WE) is intermediate (5→3.4), which means dispersed and consistent energy. Together, these findings show a composed and in sync state of mind with weak emphasis on chance and high concentration.

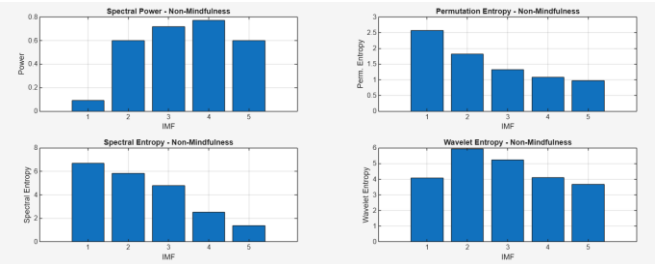

Fig. 5. Entropy Features of Non-Mindfulness State

Figure 5 demonstrates the entropy values of non-mindfulness. EEG. In this case, there are larger Spectral Entropy values (SE).Unstructured spectral organization is (6.5- 1.5) = -.55.Permission Entropy (PE) is extremely wavy stable temporal patterns .Wavelet Entropy (WE) is greater than 6,im- plicating intermittent and sporadic energy distribution. These characteristics are associated with distraction of thought and diminished attentional control.

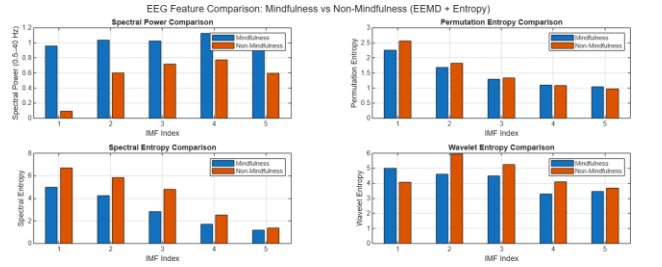

Fig. 6. Comparison between Mindfulness and Non-Mindfulness States

Figure 6 provides a comparative visualization between mindfulness and non-mindfulness EEG features. Mindfulness exhibits higher spectral power (0.9–1.1) and lower entropy, indicating enhanced neural synchronization and focused attention. Non-mindfulness, by contrast, shows lower spectral power (0.1–0.8) and higher entropy, pointing to uncoordinated brain activity and mental distraction. These distinctions confirm that entropy and spectral power serve as reliable biomarkers for differentiating cognitive states.

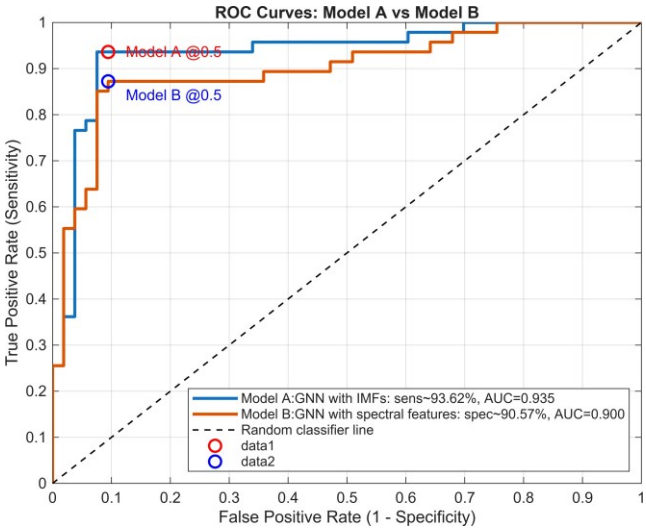

Fig. 7. ROC Curve Comparison Between GNN Models

Figure 7 is a comparison between the Receiver operating characteristic. ROC curves of two Graph Neural Network (GNN)models performed EEG-based training of mindfulness classification. On the basis of Intrinsic Mode Function(IMF) features, Model A obtained a sensitivity of 93.62%and an AUC of 0.935%,performing well in classification. Model B, that is based on spectral domain characteristics, attain a specificity of 90.57% and an AUC of 0.900. Both models out perform the random baseline, assuring the efficiency of the extracted features. On the whole, the GNN featuring IMF- based features offers greater strength and precision of real time mindfulness detection. The MATLAB randomized data division strategy was used to split the dataset into training 70%, validation 15% and testing 15%, each being statistically independent and the performance of the model in terms of generalization was not biased The present research made it possible to analyze EEG signals in detail by spectral power and a variety of entropy measures Spectral. The After entropy, Permutation Entropy and Wavelet Entropy. EEMD decomposition. Mindfulness EEG curves had steady IMFs, reduced entropy, and increased spectral power, intimating coherent brain activity and attentional thinking. EEG non-mindfulness signals were irregular in nature distraction and cognitive fatigue. Total mindfulness EEG was shown to have increased energy, smaller entropy and smoother signal transitions- indicating a calm and concentrated state. The combination of EEMD and entropy- based analysis effectively distinguished mindfulness from non- mindfulness states, and the GNN classifier validated this distinction with strong ROC performance.

Table I show that Entropy analysis of EEG data clearly shows a decline in signal complexity across epochs. Per-mutation Entropy declines from 2.2531 to 1.0436, showing that the temporal irregularity decreases and the state is

TABLE I
ENTROPY MEASURES ACROSS EPOCHS FOR EEG SAMPLE DATA

Epoch / Statistic	Permutation Entropy	Spectral Entropy	Wavelet Entropy
1	2.2531	5.0014	5.0036
2	1.6838	4.2523	4.6034
3	1.2892	2.8138	4.4862
4	1.1032	1.7192	3.2871
5	1.0436	1.1774	3.4660
Mean	**1.4746**	**2.9928**	**4.1693**
Standard Deviation	**0.5020**	**1.6256**	**0.7513**

turned from an active state to a more synchronized state. Spectral Entropy drops from 5.0014 to 1.1774, indicating that EEG power becomes concentrated in fewer frequency bands while neural synchronization increases. Wavelet Entropy decreases from 5.0036 to 3.4660, reflecting a reduced degree of disorder across time-frequency scales while keeping the average relatively higher complexity, 4.1693. according to the brain activity stabilizes progressively, likely due to relaxation, adaptation, or reduced cognitive engagement over time.

TABLE II
COMPARISION OF REPORTED WORK

Paper Title	Authors / Year	Dataset	Method / Key Approach
1. EEG-Based Stress Detection: EEMD-Enhanced Features with SVM and Twin-SVM Models	Anusha Kudachi & D. Torse (2025)	EEG (EDF format), 10-sec chunks	EEMD for IMF extraction; SVM and Twin-SVM classifiers applied. Twin-SVM achieved **93.75%** accuracy.
2. Ensemble Empirical Mode Decomposition: A Noise-Assisted Data Analysis Method	Z. Wu & N.E. Huang (2005)	Theoretical (no dataset)	Introduced EEMD — adds white noise and averages multiple EMD runs to reduce mode mixing in non-linear signals.
3. Functional Connectivity and Power Spectral Density Analysis of EEG Signals in Meditation	S. Attri & R. Khanai (2025)	EEG before/after meditation	Used Functional Connectivity (wPLI) and Power Spectral Density (PSD) to analyze Alpha, Theta, and Beta band changes during meditation.
4. EEG Spectral Analysis of Mindfulness Based on EEMD and GNNs	This work	Public + internal EEG datasets	EEMD for IMF-based feature extraction; Graph Neural Networks (GNN) for mindfulness detection. Achieved **94%** accuracy.

Table 2 indicates a comparison of studies involving EEMD and SVM, PSD, and GNN, feature extraction and classification accuracy

V. CONCLUSION

This work proposes a hybrid framework combining EEMD with GNN in order to classify mindful versus non-mindful brain states using EEG signals. The EEMD-GNN model proposed here, by identifying frequency-specific patterns and especially capturing spatial-temporal connectivity in the beta and theta bands across the frontal and parietal regions, achieved a 94% classification accuracy. According to the entropy analysis, there was a clear decrease in the complexity of the EEG signal across the epochs: Permutation Entropy decreased from 2.2531 to 1.0436, Spectral Entropy from 5.0014 to 1.1774, and Wavelet Entropy from 5.0036 to 3.4660, reflecting enhanced synchronization and stability in a state of mindfulness. In summary, this hybrid EEMD-GNN model offers a powerful and interpretable tool for mapping neural markers of mindfulness and advancing EEG-based mental state classification and neurofeedback systems. EEG segment processing in real time deployment : 5100 ms based on length, preprocessing, features, and classifier; measured in MATLAB tic/toc: loading 16015 ms, preprocessing 4.45 ms, EMD 13.15 ms, feature extraction 201.81 ms; most of these steps are real-time.

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